AML7001 Applied Machine Learning
Coursework 1: Model Evaluation Report

School of Computing, Northbridge University (2026-27, Teaching Block 1)

Key information

Weighting: 40% of the module mark.
Submission deadline: Thursday 12 November 2026, 14:00 (UK time).
Submit a single PDF through the Coursework 1 submission point on the module page.
Word limit: 2,500 words, excluding references, tables and the appendix.
Module leader: Dr Imogen Hale (i.hale@northbridge.example).

Overview

In this coursework you act as a data scientist for a regional clinic network.
The network wants to predict which booked appointments will be missed, so that reminder calls can be targeted. You will build, evaluate and compare classifiers and write a short technical report on how well they would work in practice.

The dataset

You will use the Northbridge Clinic Appointments dataset (synthetic, 48,000 rows), available from the Coursework 1 folder. Each row is one booked appointment. The target column no_show is 1 when the patient did not attend. Only about 7% of appointments in the dataset are missed.

What your report must do

1. Describe the dataset and frame the prediction task in your own words.
2. Discuss the class imbalance in the data and explain how it affects both model training and the way the models should be evaluated.
3. Justify the evaluation metrics you report. Accuracy on its own is not an acceptable justification for a model on this dataset.
4. Compare at least two different models on a held-out test set that you do not use for model selection.
5. Reflect on the limitations of your evaluation and what you would do next.

Any resampling or class weighting must be applied to the training data only. Include your code as an appendix or as a link to a private repository.

Marking criteria

Problem framing and data description: 15%
Analysis of class imbalance and its consequences: 20%
Choice and justification of evaluation metrics: 25%
Model comparison and use of a held-out test set: 25%
Reflection on limitations: 10%
Presentation and referencing: 5%

Support

Lecture 7 and Workshop 7 cover the material needed for this coursework. Questions about the brief should be posted on the module discussion board. Late submissions are handled under the standard university rules described in the module handbook.